SentinelVision AI

Volume 02

Software Requirements Specification

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> **SentinelVision AI — Enterprise Documentation Suite**

> Status legend: Implemented | Planned | Future

1. System Overview

SentinelVision AI comprises an edge inference worker, Django REST control plane, React dashboard, PostgreSQL datastore, and planned async/observability services.

1.1 Actors

Actor	Description
SecurityOperator	Monitors alerts and reviews events
FacilityAdmin	Configures cameras, sites, users
HROfficer	Manages attendance and visitors
EdgeService	Automated edge worker posting heartbeats/events
SystemAuditor	Reviews audit and compliance exports
PlatformAdmin	Operates control plane infrastructure

2. Use Cases

ID	Use Case	Primary Actor	Status
UC-AUTH-01	Authenticate with JWT	All users	Implemented
UC-CAM-01	Register camera	FacilityAdmin	Implemented
UC-EDGE-01	Edge heartbeat registration	EdgeService	Implemented
UC-EVT-01	Ingest detection event	EdgeService	Implemented
UC-EVT-02	Review detection event	SecurityOperator	Implemented
UC-ALERT-01	Acknowledge alert	SecurityOperator	Implemented
UC-ALERT-02	Configure alert rule	FacilityAdmin	Implemented
UC-ATT-01	Record attendance via face	EdgeService	Planned
UC-VIS-01	Pre-register visitor	HROfficer	Planned
UC-RPT-01	Export incident report	SystemAuditor	Planned

UC-EVT-01 Ingest Detection Event

Preconditions: Edge device authenticated; camera active.

Main flow:

1. Edge samples inference result.

2. Edge POSTs event payload to `/api/v1/events/ingest/`.
3. API validates camera ownership/site binding.
4. API persists DetectionEvent.
5. API evaluates alert rules and creates Alert records when matched.

Alternate flows:

- A1: Duplicate `edge_event_id` returns existing event (200).
- A2: Invalid camera returns 404.
- A3: Camera outside edge site returns 403.

Postconditions: Event stored; optional alert created.

UC-EVT-02 Review Detection Event

Preconditions: Operator authenticated with Operator+ role.

Main flow:

1. Operator opens event review queue.
2. Operator selects pending event.
3. Operator marks confirmed or false positive.
4. System records reviewer and timestamp.

3. UML Use Case Diagram

```
flowchart LR
subgraph actors [Actors]
Operator[SecurityOperator]
Admin[FacilityAdmin]
Edge[EdgeService]
HR[HROfficer]
end
subgraph system [SentinelVision]
UC1[ManageCameras]
UC2[IngestEvents]
UC3[ReviewEvents]
UC4[ManageAlerts]
UC5[ManageVisitors]
end
Admin --> UC1
Edge --> UC2
Operator --> UC3
Operator --> UC4
HR --> UC5
```

4. Activity Diagram — Event Processing

1. Edge captures frame
2. Detector produces candidates
3. Module router evaluates zones/rules
4. Event payload assembled
5. HTTP POST with retry
6. API validation
7. Persist event
8. Evaluate alert rules
9. Dashboard refresh

5. Sequence Diagram — Event Ingestion

```
sequenceDiagram
participant Edge
participant API
participant DB
participant Rules as AlertRules
Edge->>API: POST /events/ingest
API->>DB: Validate camera
API->>DB: Insert DetectionEvent
API->>Rules: evaluate_alert_rules
Rules->>DB: Insert Alert optional
API-->>Edge: 201 Created
```

6. State Diagram — Alert Lifecycle

State	Description
open	Created, awaiting operator action
acknowledged	Operator acknowledged
resolved	Incident closed
dismissed	False alarm or non-actionable

Transitions: open→acknowledged (operator), acknowledged→resolved (operator), open→dismissed (operator/admin).

7. Business Rules

ID	Rule
BR-TEN-01	All user queries must be scoped to member organizations
BR-RBAC-01	Viewer cannot mutate configuration
BR-EDGE-01	Edge may ingest only for cameras on its site
BR-ALERT-01	Alert severity must be >= rule min severity
BR-ALERT-02	Event confidence must be >= rule min confidence
BR-BIO-01	Face enrollment requires consent record (Planned)
BR-RET-01	Evidence retention defaults to 30 days (Planned enforcement)

8. Validation Rules

Entity	Field	Validation
Camera	name	Required, unique per site
Camera	rtsp_url	Valid URL format
Zone	polygon_json	Array of normalized [x,y] pairs
DetectionEvent	confidence	0.0–1.0
AlertRule	min_confidence	0.0–1.0
EdgeDevice	api_key	One-time display on create

9. Error Handling

Code	Condition	Response
400	Serializer validation failure	DRF validation errors
401	Missing/invalid JWT or edge key	Authentication failed
403	Role or site mismatch	Permission denied
404	Resource not found	Not found
409	Planned idempotency conflict	Future standardized envelope

Target error envelope (Planned):

```
{"error": {"code": "VALIDATION_ERROR", "message": "...", "details": {}}}
```

10. Workflow Specifications

10.1 Human Review Workflow

Pending → Confirmed | False Positive | Escalated

Escalated events require supervisor review (Planned).

10.2 Edge Offline Workflow

1. Heartbeat missed beyond threshold
2. Edge marked offline (Planned automated rule)
3. Cameras at site marked degraded
4. Operator notification (Planned)

11. Security Requirements

- JWT access token 15 minutes; refresh 7 days
- Edge API keys stored hashed (SHA-256)
- Tenant isolation on all list/detail endpoints
- Planned: rate limiting, key rotation, constant-time compare

12. Traceability

Each UC maps to FR IDs in Volume 1 and API IDs in Volume 5.